DISCRETE STRUCTURES

CMSC250

Chapter 22: Probability

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SUBProbability



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1 Why probability builds on counting

Probability quantifies uncertainty about outcomes. In a *finite equally likely* sample space Ω, the probability of an event $A \subseteq \Omega$ is

$$P(A) = \frac{|A|}{|\Omega|}.$$

This counting formula is valid only when every elementary outcome in Ω has the same probability. Always describe the experiment and its outcomes before counting.

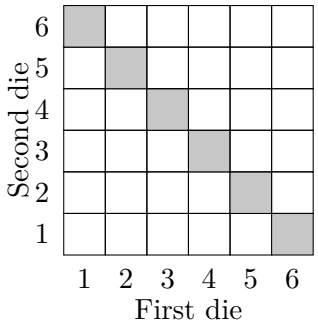
2 Sample spaces and events

A *sample space* Ω contains all possible outcomes of an experiment. An *event* A is a subset of Ω. For one fair six-sided die, $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the event of an even roll is $E = \{2, 4, 6\}$. Thus $P(E) = 3/6 = 1/2$.

Worked example: two fair dice

Question. What is the probability that two independent fair dice sum to seven?

Solution. The sample space consists of 36 equally likely *ordered pairs* (i, j), where $i, j \in \{1, \ldots, 6\}$. The favorable pairs are $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$, so the probability is $6/36 = 1/6$.



The shaded squares represent the six outcomes whose sum is seven.

3 Basic probability laws

For any event A, $0 \leq P(A) \leq 1$, $P(\Omega) = 1$, and $P(\varnothing) = 0$. The complement $A^c = \Omega \setminus A$ satisfies $P(A^c) = 1 - P(A)$. For any events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are disjoint, then $P(A \cap B) = 0$ and their union probability is the sum.

Worked example: at least one six

Question. Two independent fair dice are rolled. What is the probability of at least one six?

Solution. The complement is *neither die shows six*. Its probability is $(5/6)^2 = 25/36$. Therefore $P(\text{at least one six}) = 1 - 25/36 = 11/36$.

4 Conditional probability

For events A, B with $P(B) > 0$, the probability of A *given* B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Conditioning changes the relevant universe: once B is known to occur, outcomes outside B are excluded.

Worked example: conditional probability with a die

Question. Given that a fair die shows an even number, what is the probability it is greater than three?

Solution. Let $B = \{2, 4, 6\}$ and $A = \{4, 5, 6\}$. Then $A \cap B = \{4, 6\}$. Therefore $P(A \mid B) = (2/6)/(3/6) = 2/3$.

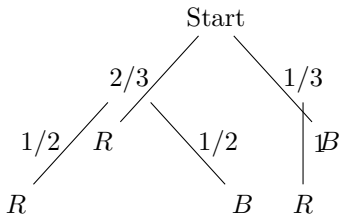
5 Probability trees

A probability tree labels each branch with the probability of that branch *conditional on reaching its parent node*. Multiply along a path to obtain the probability of that complete outcome. Add probabilities of disjoint paths to obtain an event probability.

Worked example: two draws without replacement

Question. A bag contains two red balls and one blue ball. Two balls are drawn without replacement. What is the probability of drawing two reds?

Solution. The first red has probability $2/3$. Given a red first, the second red has probability $1/2$. Therefore $P(RR) = (2/3)(1/2) = 1/3$.



6 Multiplication rule

Rearranging the definition of conditional probability yields

$$P(A \cap B) = P(B)P(A \mid B) = P(A)P(B \mid A),$$

whenever the relevant conditional probability is defined. This is valid even when the events are dependent.

7 Independent events

Events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B).$$

When $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$: learning that B occurred does not change the probability of A.

Worked example: independence of two dice

Question. Let A be the event that the first die is even and B the event that the second die is six. Are A and B independent?

Solution. There are 18 outcomes in A, six in B, and three in $A \cap B$. Thus $P(A) = 1/2$, $P(B) = 1/6$, and $P(A \cap B) = 3/36 = 1/12$. Since $(1/2)(1/6) = 1/12$, the events are independent.

8 Disjoint is not the same as independent

Question. Prove that two disjoint events with positive probability cannot be independent.

Proof. Let A and B be disjoint, with $P(A) > 0$ and $P(B) > 0$. Since $A \cap B = \varnothing$, we have $P(A \cap B) = 0$. But $P(A)P(B) > 0$ because both factors are positive. Therefore $P(A \cap B) \neq P(A)P(B)$, so A and B are not independent. ■

Why the proof works. Disjoint events exclude one another; independent events do not affect one another's probabilities.

9 Counting probability with combinations

When selecting a uniformly random r-element subset of an n-element set, the sample space has $\binom{n}{r}$ equally likely subsets.

Worked example: drawing two aces

Question. Two cards are drawn uniformly without replacement from a standard 52-card deck. What is the probability that both are aces?

Solution. There are $\binom{52}{2}$ unordered two-card hands, of which $\binom{4}{2}$ contain two aces. Hence

$$P(\text{two aces}) = \frac{\binom{4}{2}}{\binom{52}{2}} = \frac{6}{1326} = \frac{1}{221}.$$

10 Bayes' theorem

For $P(B) > 0$, Bayes' theorem follows from the multiplication rule:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

When A and A^c partition the sample space, the denominator may be written

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

Worked example: a diagnostic test as a probability problem

Question. A condition affects 1% of a population. A test is positive for 90% of people with the condition and 5% without it. What is the probability a randomly chosen person has the condition, given a positive result?

Solution. Let C mean condition and $+$ mean positive. Then

$$P(C \mid +) = \frac{0.90(0.01)}{0.90(0.01) + 0.05(0.99)} = \frac{0.009}{0.0585} \approx 0.154.$$

The positive predictive probability is about 15.4%, not 90%. The base rate matters.

11 Practice questions and complete solutions

Question. 1. Three fair coins are tossed. What is the probability of exactly two heads?

Solution. There are $2^3 = 8$ equally likely ordered outcomes and $\binom{3}{2} = 3$ positions for the two heads. The probability is 3/8. **Question.** 2. A fair die is rolled twice. What is the probability the second result exceeds the first?

Solution. Of the 36 ordered pairs, six are ties. The remaining 30 split equally between second greater and first greater by symmetry. Hence the probability is $15/36 = 5/12$. **Question.** 3. A box contains three red and two blue balls. Two are drawn without replacement. Find $P(\text{second red} \mid \text{first blue})$.

Solution. After a blue is drawn, four balls remain, three red and one blue. Thus the conditional probability is 3/4. **Question.** 4. If $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$, are the events independent?

Solution. Yes, because $P(A)P(B) = 0.4 \cdot 0.5 = 0.2 = P(A \cap B)$.

12 Common mistakes

Do not assume outcomes are equally likely without justification. Two dice have 36 ordered outcomes, not 11 equally likely sums. Do not confuse $P(A \mid B)$ with $P(B \mid A)$. When sampling without replacement, update the denominator after each draw. Disjoint events with positive probabilities are never independent.